

The Program in Action

When you run *waterbucket.py*, the output will look like this:

Water Bucket Puzzle, by Al Sweigart al@inventwithpython.com

Try to get 4L of water into one of these buckets:

```
8|      |
7|      |
6|      |
5|      | 5|      |
4|      | 4|      |
3|      | 3|      | 3|      |
2|      | 2|      | 2|      |
1|      | 1|      | 1|      |
+-----+ +-----+ +-----+
      8L      5L      3L
```

You can:

- (F)ill the bucket
- (E)mpty the bucket
- (P)our one bucket into another
- (Q)uit

```
> f
Select a bucket 8, 5, 3, or QUIT:
> 5
```

Try to get 4L of water into one of these buckets:

```
8|      |
7|      |
6|      |
5|      | 5| WWWWWW|
4|      | 4| WWWWWW|
3|      | 3| WWWWWW| 3|      |
2|      | 2| WWWWWW| 2|      |
1|      | 1| WWWWWW| 1|      |
+-----+ +-----+ +-----+
      8L      5L      3L
--snip--
```

How It Works

The `waterInBucket` variable stores a dictionary that represents the state of the water buckets. The keys to this dictionary are the strings `'8'`, `'5'`, and `'3'` (representing the buckets), and their values are integers (representing the liters of water in that bucket).

Lines 48 to 59 use this dictionary to render the buckets and water on the screen. The `waterDisplay` list contains either `'WWWWWW'` (representing water) or `' '` (representing air) and is passed to the `format()` string